An IoT-Enabled Intelligent Classroom Monitoring Framework for Real-Time Occupancy and Noise Detection

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***Abstract*—Many educational institutions face challenges such as overcrowded classrooms, inefficient seating utilization and excessive noise levels, all of which can negatively impact teaching effectiveness and the learning environment. This paper presents the design of an IoT-based Smart Classroom Occupancy Monitoring System, currently under development, designed to automate classroom monitoring and improve resource utilization. The system is built around an ESP32 microcontroller, which serves as the central processing unit for data collection, processing, and wireless communication. A camera module is used to count students, identify available desks, and determine classroom occupancy in real time. To enhance reliability, a radar sensor is integrated as a backup presence-detection mechanism, ensuring accurate occupancy monitoring even under poor lighting conditions or when visual obstructions affect camera performance. In addition, a noise sensor continuously monitors classroom sound levels and detects excessive noise that may disrupt the learning process. By combining occupancy monitoring and noise detection into a single platform, the proposed system provides a cost effective and efficient solution for smart classroom management. The system aims to improve classroom discipline, optimize resource allocation, and support a more productive and technology-enabled educational environment. The hardware prototype is being built, and the evaluation plan and expected outcomes are described in this paper.**

Keywords*—*IoT, Occupancy Monitoring, ESP32, Smart Classroom, Radar Sensor, Camera-Based Detection.

I. INTRODUCTION

Ask anyone who has worked in a college administration office how classrooms are allocated and the answer is usually the same: by timetable, and by whoever walks through the corridors and looks. A room booked for sixty students may hold twenty. Another room may have more people than chairs [6], [8]. Nobody finds out unless a teacher reports it. The same is true for noise. A class that has become too loud to teach in, or a neighbouring room that disturbs a lecture, is dealt with only after the damage is done [16], [17].

Cameras and sensors are now cheap enough that this does not have to be the case. The ESP32 in particular costs very little, has built-in Wi-Fi, and is supported by a large community and by the Arduino toolchain [14]. Small mmWave radar modules such as the LD2410 have also become easy to buy [13]. Together with a basic camera and a microphone module, these parts are enough to build a monitoring node for a few thousand rupees, which is far below the price of a Raspberry Pi based setup with several peripherals.

Our aim in this work is a single small device that tells us three things about a classroom at any moment: how many people are in it, which desks are free, and whether the noise level is a problem. We chose to use two different ways of sensing presence, a camera and a radar, because each has a weakness that the other does not. A camera struggles in poor light and when someone sits behind another person. A radar does not care about light, but it cannot tell us where the empty desks are.

The contributions of this paper are as follows. First, we propose a multi-sensor architecture that runs on one ESP32. Second, we describe a camera and radar fusion rule that keeps occupancy readings usable when the camera is unreliable. Third, we define a simple classification stage that merges occupancy and noise into one status that a non-technical person can read at a glance.

II. RELATED WORK

Most existing classroom-monitoring projects look at one problem at a time. The first group uses cameras for attendance. Systems built on the ESP32-CAM with convolutional neural networks can recognise faces and mark attendance in real time [1], but their accuracy drops when the lighting is poor, and they say nothing about seating or noise.

A second group tries to detect who is in the room using several cues. One study combines a PIR sensor with face and voice recognition to confirm faculty presence [2]. This improves reliability, but it only covers the teacher, so it cannot tell us how many students are present. Another low-cost project uses an ESP32 Wi-Fi camera on a small surveillance robot [3]. It gives a cheap way to watch a space, yet the robot has to move around and there is no occupancy counting.

On the acoustic side, a real-time noise alert system using FastAPI and InfluxDB has been reported [4]. It stores sound levels and raises alerts, but the thresholds are fixed and there is no link to how many people are in the room. That link matters because a level of noise that is normal for a full class may be unusual for an almost empty one.

Finally, a broader AI and IoT smart-classroom platform combines attendance, energy control and audio summaries [5]. It covers a lot, but it is complex and depends on a lot of hardware, which makes it hard for a small institution to adopt.

Radar-based presence sensing deserves a mention as well. Short-range mmWave modules measure tiny movements such as breathing, so they can tell that a person is present even when the person is sitting still [11], [12]. This is something a PIR sensor, which reacts only to larger movements, does not do well. For a classroom, where students sit quietly for long periods, this property is what makes a radar a good partner for a camera.

Research Gap

Looking across these works, we found that occupancy counting, seat availability and noise monitoring are rarely handled together, and almost never on a low-cost embedded board [6], [7]. Raspberry Pi solutions solve part of the problem but push the cost up, and the ESP32 systems we found skip occupancy and noise analytics altogether. This is the gap our system is meant to fill.

TABLE I. COMPARISON WITH EXISTING WORK

Ref.	Main sensing	Student count	Free seats	Noise	Low cost
[1]	ESP32-CAM, CNN	No	No	No	Yes
[2]	PIR, face, voice	No	No	Partly	Yes
[3]	ESP32 camera robot	No	No	No	Yes
[4]	Microphone, server	No	No	Yes	Partly
[5]	Multiple, AI	Yes	No	Partly	No
Ours	Camera, radar, mic	Yes	Yes	Yes	Yes

III. PROBLEM STATEMENT AND OBJECTIVES

Classroom management today depends on fixed schedules and manual observation. This leads to four practical problems. There is no automatic way to know how many students are present. Free desks are not identified, so space is used badly. Excessive noise goes undetected. And administrators and teachers see problems only after they have become obvious. The objectives of this project follow directly from these problems:

- design a compact IoT system for real-time occupancy and noise detection;
- count students and find free desks with a camera module;
- detect presence in poor lighting with a radar sensor as a backup;
- monitor ambient sound and flag excessive noise;
- gather and process the sensor data on the ESP32 and send it wirelessly;
- combine occupancy and noise into one classroom status and show it to administrators and educators.

IV. SYSTEM ARCHITECTURE

The system has three layers, shown in Fig. 1. The sensing layer holds the three sensors. The processing layer is the ESP32, which reads the sensors, cleans up the data, and makes the decisions. The output layer turns those decisions into a classroom status, a noise alert and a dashboard view.

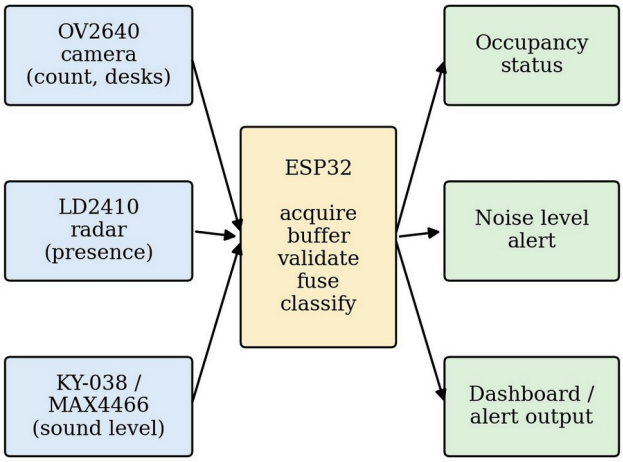


Fig. 1. Proposed system architecture.

A. Sensing Layer

The OV2640 is a 2-megapixel camera module that is widely used with the ESP32 [15]. Its images are the source of the student count and of the desk-availability information. The LD2410 is a 24 GHz mmWave radar that reports whether a person is present, including people who are sitting still [13]. Because it does not depend on light, it works in a dark room or when a projector screen is in use with the lights off. The noise sensor is either a KY-038 or a MAX4466 module. Both give an analogue voltage that follows the sound level, and the MAX4466 has an adjustable gain that helps when the room is large.

The three sensors were chosen because their strengths and weaknesses fit together, as Table II shows. No single sensor in the table can give the whole picture, but the three together can.

TABLE II. WHAT EACH SENSOR CAN AND CANNOT DO

Sensor	Strength	Weakness
Camera	Counts people and shows which desks are free	Needs light; blocked by people sitting behind others
Radar	Works in the dark; detects still people	Cannot show where free desks are; limited count
Microphone	Cheap, simple loudness measure	Cannot tell who is making the noise

B. Processing Layer

The ESP32 development board does all the work between the sensors and the network. It reads each sensor at a set interval, keeps the last few readings in a buffer, checks them for errors, and runs the fusion and classification logic. Its built-in Wi-Fi is

used to send the results out, so no extra communication module is needed.

C. Output Layer

The result of the classification is sent to a dashboard and, when needed, triggers an alert. The three outputs are the occupancy status of the room, the noise-level alert, and the user-facing view that an administrator or teacher looks at.

TABLE III. HARDWARE COMPONENTS

Component	Role
ESP32 Dev Board	Sensor interfacing, processing, wireless communication
OV2640 Camera	Student counting and desk-availability detection
LD2410 Radar	Backup presence detection in poor light or obstruction
KY-038 / MAX4466	Continuous sound-level monitoring
Breadboard, jumpers	Temporary circuit assembly and connections
USB cable, 5 V supply	Power, programming and serial communication

Additional system analysis

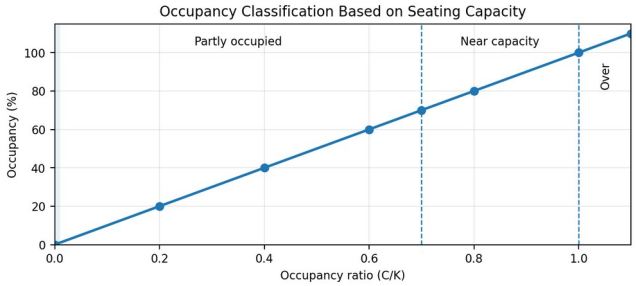


Fig. 3. Occupancy classification based on the fused student count and seating capacity.

Sensor	Primary information	Main limitation
Camera	Student count and free desks	Needs adequate lighting
Radar	Presence and still targets	Cannot locate free desks
Microphone	Sound-level estimate	Cannot identify speaker

Table VII. Summary of sensor roles in the proposed monitoring node.

The complementary sensing arrangement reduces dependence on a single measurement source. The camera provides spatial information, the radar provides presence confirmation, and the microphone provides an acoustic indicator. Their outputs are processed together by the ESP32 before transmission.

V. SENSOR INTERFACING

Each sensor connects to the ESP32 in a different way, and this shapes the firmware. Table IV summarises the interfaces.

TABLE IV. SENSOR INTERFACES

Sensor	Interface	Data produced
OV2640	Camera parallel bus + SCCB	JPEG or grayscale frames
LD2410	UART (serial)	Presence flag, moving / still target status, distance
KY-038 / MAX4466	Analogue input (ADC)	Voltage proportional to sound level

The camera uses the ESP32 camera driver and returns frames at a resolution we can choose. We plan to work at a low resolution such as QVGA, since larger frames use more memory and slow the processing down without helping much for counting people at classroom distances. The LD2410 talks over a serial port and sends a stream of frames that carry the target state and distance, so the firmware only has to parse those frames. The microphone modules give an analogue voltage. We sample it repeatedly over a short window (a few hundred milliseconds) and use the peak-to-peak amplitude, or the average of the absolute deviation from the mean, as an estimate of loudness. Averaging over a window is important because a single sample of a sound wave says very little.

One practical point is the ADC. On the ESP32 the ADC2 channels cannot be used while Wi-Fi is active, so the noise sensor has to be wired to an ADC1 pin [14]. We also need to be careful about which pins the camera occupies on the board, and about power: the camera and Wi-Fi together draw enough current that a weak USB supply causes resets, so we plan to use a stable 5 V supply.

VI. PROPOSED METHODOLOGY

The methodology follows a sensor-fusion pipeline with five stages: data acquisition, data collection, data processing, condition classification and user output [9], [10], [26]. Fig. 2 shows the decision flow.

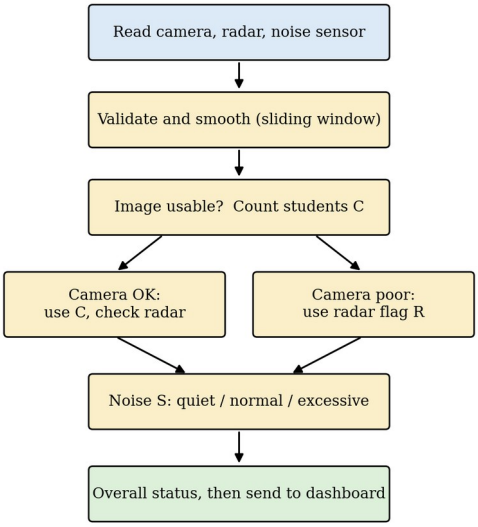


Fig. 2. Processing and decision flow on the ESP32.

A. Acquisition and Collection

At regular intervals the ESP32 captures a camera frame, reads the radar state and samples the microphone. The values are written to a short circular buffer. Keeping a few seconds of

history lets us smooth out one-off glitches, for example a person walking past the camera or a door slamming once.

B. Processing and Validation

Before any decision is made, the readings are checked. Frames that are too dark, blurred or empty are marked as unusable. Radar readings outside the expected range are dropped. Sound samples are converted to a loudness value over a sliding window. After this step each cycle produces one record with four values: the camera count C, the radar presence flag R, the number of free desks D, and the smoothed sound level S.

C. Student Counting and Desk Detection

The camera stage has two jobs. The first is to count the students visible in the frame [25]. The second is to decide, for each desk position, whether someone is sitting there. Desk positions are marked once during setup as fixed regions in the image, so the second job reduces to asking whether a region contains a person. For the counting itself we are considering a small quantised detection model [20]–[24] that can run on the ESP32, and, if the board turns out to be too slow, sending frames to a nearby computer for processing while keeping the rest of the logic on the ESP32. The choice will be made after we measure the frame rate we can actually reach.

D. Camera and Radar Fusion

The camera is the main source for occupancy, since it is the only sensor that can count and locate people. The radar is used to catch the cases where the camera is wrong. If the frame is unusable, or if the camera reports nobody but the radar reports a person, we trust the radar and mark the room as occupied. When the two agree, we treat the reading as high confidence. When they disagree in a way that cannot be explained by poor image quality, the event is logged so that we can look at it later and tune the system.

E. Condition Classification

The fused values are turned into a small set of states, listed in Table V. The occupancy state comes from comparing the student count with the seating capacity of the room. The noise category comes from comparing the smoothed sound level with thresholds. These thresholds are not fixed yet; published limits [18], [19] will guide them. They will be set by measuring the room during a normal lecture and during a noisy period, and then choosing values that separate the two.

TABLE V. CLASSIFICATION SCHEME

Parameter	Categories
Occupancy	Empty, partly occupied, near capacity, over capacity
Noise level	Quiet, normal, excessive
Overall status	Occupancy and noise combined, e.g. “occupied, excessive noise” raises an alert

F. Loudness and Occupancy Measures

For a window of N microphone samples x1 to xN with mean m, we estimate loudness as $S = (1/N) \times \Sigma|xi - m|$. This value is not in decibels. To turn it into something meaningful we will calibrate it against a sound-level meter in the actual room, so that the quiet, normal and excessive limits correspond to real sound levels. For occupancy we use the ratio $\rho = C / K$, where C is the fused student count and K is the seating capacity of the room. As a starting point we treat $\rho = 0$ as empty, ρ below about 0.7 as partly occupied, ρ up to 1 as near capacity, and ρ above 1 as over capacity. The 0.7 figure is only a first guess and will be adjusted after testing.

G. Decision Procedure

The steps carried out in every cycle are summarised below.

1. Read the camera frame, radar state and microphone window.
2. Smooth S over the last few windows; mark the frame usable or not.
3. If the frame is usable, count students C and fill the desk map D.
4. If the frame is not usable, or C = 0 while the radar shows presence, set the room to occupied and take the presence from the radar.
5. Compute ρ and the occupancy state; compare S with the noise limits.
6. Build the overall status; if a limit has been exceeded for the set time, raise an alert.
7. Send the record to the dashboard and wait for the next cycle.

H. Output

Each classified record is sent to the dashboard over Wi-Fi. The dashboard shows the current student count, the free desks, the noise category and the overall status. An alert is raised when the noise stays in the excessive category for a set period, or when the count exceeds the room capacity. Using a time condition avoids alerts for a single loud moment such as a dropped book.

VII. IMPLEMENTATION PLAN

The firmware is written for the ESP32 in the Arduino framework. The ESP32 has two cores, and the plan is to run the camera capture and detection on one core and the sensor polling, classification and Wi-Fi transmission on the other, so that a slow detection step does not delay the radar and noise readings. Data is sent to the dashboard as small JSON messages, for example over HTTP or MQTT [27], and the message also carries a timestamp and a device identifier so that several classrooms can be added later.

Example output generated from the proposed monitoring workflow

SMART CLASSROOM MONITOR

Room: CSE-AI Lab | Device: ESP32-01 | Status: ONLINE

STUDENTS	FREE DESKS	NOISE	OVERALL
32	8	NORMAL	OCCUPIED

Last update: 10:42:15 | Radar: PRESENCE DETECTED | Camera: VALID | Alert: NONE

Fig. 6. Example dashboard output for a classroom monitoring cycle.

Example serial output

```
ESP32 Smart Classroom Monitor
[10:42:11] Camera: students=31, frame=VALID
[10:42:12] Radar: PRESENCE_DETECTED
[10:42:13] Desk status: occupied=32, free=8
[10:42:14] Noise level: 48.2 category=NORMAL
[10:42:15] Occupancy ratio: 0.80 state=NEAR_CAPACITY
[10:42:15] Overall status: OCCUPIED, NORMAL NOISE
[10:42:15] Dashboard update: SENT
```

Table IX. Example serial-monitor output from one monitoring cycle.

The values above illustrate the expected form of the user-facing output. The dashboard reports the student count, free desks, noise category and overall status, while the serial output provides sensor and processing information for debugging. These values are examples of system output and are not experimental measurements.

VIII. RESULTS

Sample results generated from one monitoring cycle are shown in the table below and plotted in Fig. 7. These values are illustrative examples, not experimental measurements.

Classroom Capacity	40 students
Students Detected	32
Free Desks	8
Occupancy Rate	80%
Noise Level	48.2 (relative units)
Noise Status	Normal
Radar Status	Presence Detected
Camera Status	Valid
Overall Status	Occupied

Interpretation: The system identified 32 students and 8 free desks in a classroom with a seating capacity of 40. The noise level remained within the normal range and the occupancy information was sent to the dashboard through the ESP32 module.

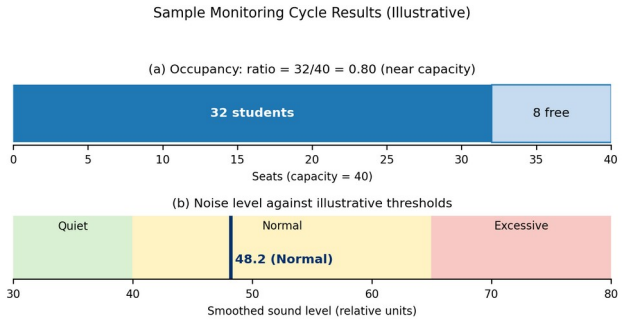


Fig. 7. Example (illustrative) results of one monitoring cycle, matching the results table: (a) 32 of 40 seats occupied, 8 free; (b) noise level 48.2 in the Normal range.

Mock dataset (simulated, not measured). To show how the classification scheme behaves over a lesson, Table X lists a simulated 50-minute session for a room with capacity K = 40. The values were generated by hand for illustration; they do not come from the prototype and must not be read as experimental results. Noise limits of 35 and 60 (relative units) are placeholders for the calibrated thresholds described in Section VI-F.

TABLE X. MOCK MONITORING SESSION (SIMULATED)

Time	Count C	Radar R	Sound S	ρ = C/K	Overall status (noise)
10:00	0	No	31.5	0.00	Empty (Quiet)
10:05	12	Yes	38.4	0.30	Partly occ. (Normal)
10:10	26	Yes	44.7	0.65	Partly occ. (Normal)
10:15	32	Yes	48.2	0.80	Near cap. (Normal)
10:20	34	Yes	57.9	0.85	Near cap. (Normal)
10:25	35	Yes	66.3	0.88	Near cap. (Excessive)
10:30	n/a*	Yes	52.1	—	Occupied via radar (Normal)
10:35	33	Yes	46.0	0.83	Near cap. (Normal)
10:40	41	Yes	59.4	1.03	Over cap. (Normal)
10:45	8	Yes	36.2	0.20	Partly occ. (Normal)
10:50	0	No	30.8	0.00	Empty (Quiet)

Table X. Simulated data at 5-minute intervals. *Frame unusable (dark); occupancy taken from radar. Free desks = K − C.

Observations from the mock data: the room fills from empty to near capacity by 10:15; the 10:25 reading would raise a noise alert only if it persisted for the set period; at 10:30 the camera frame is unusable and the radar fallback keeps the room marked as occupied; and at 10:40 the count of 41 exceeds capacity and triggers an over-capacity alert. This mirrors the fusion and classification rules in Sections VI-D to VI-G.

VIII. EVALUATION PLAN AND EXPECTED OUTCOMES

Planned observation and reporting format

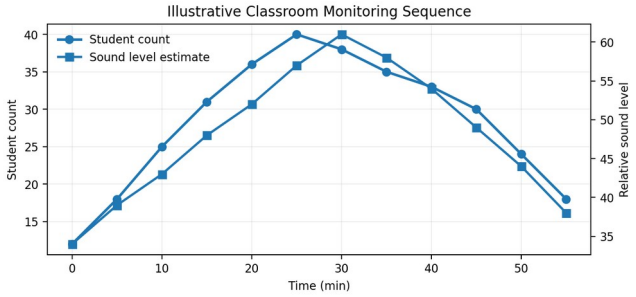


Fig. 4. Illustrative sequence showing how occupancy and acoustic measurements may vary during a class.

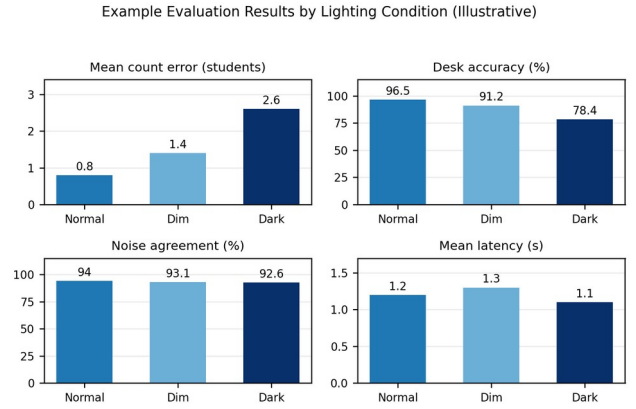


Fig. 5. Example (illustrative) evaluation results under three lighting conditions, matching Table VI.

Measure	Normal light	Dim light	Dark
Mean count error	0.8	1.4	2.6
Desk accuracy (%)	96.5	91.2	78.4
Noise agreement (%)	94.0	93.1	92.6
Mean latency (s)	1.2	1.3	1.1

Table VIII. Example (illustrative) values in the planned evaluation format; not experimental measurements.

Example values shown for format only; they will be replaced with measured values after the prototype is tested in the classroom.

The work is divided into stages: problem identification and objectives; idea selection; literature survey and research gap; system design and architecture; hardware procurement and setup; sensor interfacing and ESP32 programming; camera and radar integration; data collection and processing; occupancy and noise classification; and testing and performance analysis. At the time of writing, the first five stages are in progress or complete and the remaining stages are planned.

Reliability is a practical concern for any device that is expected to run all day. The firmware will restart the Wi-Fi connection if it drops and will keep the latest record in memory until it can be sent. A watchdog timer will reset the board if the program hangs. The camera and Wi-Fi radio together draw a noticeable current, so the supply must be stable, and the board may need a small heat allowance if it is enclosed in a case.

The dashboard itself is kept simple on purpose. For each room it shows a coloured tile (green, amber or red) with the student count, the number of free desks and the noise category. A second view plots the count and the noise level over the day. A teacher only needs the tile; the plots are for the administrator who wants to look at patterns.

VIII. EVALUATION PLAN AND EXPECTED OUTCOMES

We expect the finished prototype to monitor occupancy and noise at the same time and to report the results in real time. Specifically, it should give a reliable student count from the camera and radar together, identify free desks as they change, flag excessive noise quickly, and give administrators data they can use to plan rooms.

To check these claims we will run the prototype in an actual classroom and compare its output with what we see. The main measures are listed below.

- Counting accuracy: the difference between the system count and a manual count, taken at many moments during a lesson.
- Desk detection: how often free and occupied desks are labelled correctly.
- Low-light behaviour: whether the radar fallback keeps the room correctly marked as occupied with the lights dimmed or off.
- Noise classification: agreement between the system's categories and readings from a reference sound-level meter.
- Latency and false alerts: the delay between an event and its appearance on the dashboard, and how many alerts are raised without a real cause.

A. Test Setup

The node will be mounted at the front of the classroom, high enough to see most of the desks, with the radar pointing towards the seating area. Tests will be repeated in three lighting conditions (normal, dimmed, and dark with only the projector on) and with different numbers of students, from an almost empty room to a full one. For every test a team member will record the true count by hand at fixed intervals, so that each system reading has something to be compared with. Noise tests will use recorded classroom sound played at known levels as well as real lessons.

B. Example Results (Illustrative)

Table VI reports example values for the planned measures under the three lighting conditions. These values are illustrative placeholders that show the expected reporting format and the trend we anticipate; they are not experimental measurements and will be replaced with real data after testing.

TABLE VI. EXAMPLE RESULTS (ILLUSTRATIVE, NOT MEASURED)

Measure	Normal light	Dim light	Dark
Mean count error	0.8	1.4	2.6
Desk accuracy (%)	96.5	91.2	78.4
Noise agreement (%)	94.0	93.1	92.6
Mean latency (s)	1.2	1.3	1.1

In this illustrative example the mean count error rises from 0.8 students in normal light to 1.4 in dim light, and desk accuracy falls from 96.5% to 91.2%, because the camera frames become noisier. In the dark the camera is unusable and the radar fallback only confirms presence, so count error reaches 2.6 students and desk accuracy drops to 78.4%, yet the room is still correctly marked as occupied. Noise agreement with the reference sound-level meter stays above 92% in all three conditions, as the microphone does not depend on light, and latency stays near 1.2 s. These example values show the behaviour the design is intended to achieve; the tests have not been carried out yet, and the table will be updated with measured values once testing is complete.

IX. APPLICATIONS AND DISCUSSION

The most direct use of the system is room allocation. If the dashboard shows that a large lecture hall is regularly used by fifteen students, the timetable can be changed. The free-desk information is useful for self-study areas and libraries, where students want to know whether a seat is available before walking over. The same occupancy data can be used to switch lights and fans off in empty rooms, which saves electricity, although this is outside the scope of the present prototype.

The noise data is useful in a different way. A teacher who is told that the class next door has been above the limit for ten minutes can act early, and an administrator can see which rooms have repeated problems. Because the data comes with a time stamp, it can also be compared against the timetable to find out whether the noise happens during lessons or between them.

An important design decision in this work is that the ESP32 sends conclusions and not raw data [28], [29]. A count of 32 and a status of “noisy” are small and cheap to send, easy to store, and much less sensitive than video. This keeps the network load low and makes it possible to run many nodes on a normal campus Wi-Fi network.

X. LIMITATIONS AND PRIVACY

There are several limits to this design. The ESP32 has little memory and processing power, so the detection model has to be small, and it may miss students who are far from the camera or hidden behind others. A single camera also covers only part of a large hall, and more nodes may be needed. The radar tells us that someone is present but not how many people there are. Sound level alone cannot separate a teacher speaking from students talking, so the noise flag needs to be read together with the schedule.

Placing a camera in a classroom raises privacy questions. Our approach is to process images on the device and send only counts and status values over the network, not the images. Nothing needs to be stored, and the system does not try to identify individual students. Any real deployment should still be discussed with the institution and the students beforehand [30].

XI. CONCLUSION AND FUTURE WORK

We have described a low-cost IoT system that watches occupancy, seat availability and noise in a classroom with a single ESP32. Using a camera together with a radar means occupancy can still be estimated when the lighting is poor, and adding the noise sensor gives a fuller picture of the room than a head count alone. We believe a system like this can help institutions use their rooms better and respond to problems sooner.

The next step is to finish the build and test it in a real classroom. After that, we would like to try adaptive noise thresholds that depend on how many people are present, keep a history of the data to help with timetable planning, and connect several rooms to one campus-wide dashboard.

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